



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)

Mid-Semester Examination

Subject: Advanced Quantum Mechanics 1

Subject Code(s): (PHS 4102)

Full marks: 25

Time allotted: 2.0 hrs

1(a) Show

$$(\sigma \cdot \mathbf{a})(\sigma \cdot \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} + i\sigma \cdot (\mathbf{a} \times \mathbf{b})$$

where σ are the Pauli matrices and $\mathbf{a}, \mathbf{b}$ are vectors.

1(b) Use the above identity and show that :

$$U(\hat{n}, \phi) \equiv \exp - i \left(\frac{\sigma \cdot \hat{n}}{2} \phi \right) = I \cos(\phi/2) - i\sigma \cdot \hat{n} \sin(\phi/2)$$

1(c) Obtain

$$D_{nm'}^{1/2}(\phi, \hat{n})$$

7

2. Calculate the CG coefficients needed to couple the two angular momenta $j_1 = \frac{1}{2}$ and $j_2 = \frac{1}{2}$ to the possible j values, and express the coupled states $|j_1 j_2 j m\rangle$ to the uncoupled basis $|j_1 j_2 m_1 m_2\rangle$.

7

3. Construct a spherical tensor of rank 1 out of two different vectors $\mathbf{U} = (U_x, U_y, U_z)$ and $\mathbf{V} = (V_x, V_y, V_z)$. Explicitly write $T_{\pm 1, 0}^{(1)}$ in terms of $U_{x,y,z}$ and $V_{x,y,z}$. ***

5

4(a) A particle in a central potential has an orbital angular momentum $l = 2\hbar$ and a spin $s = 1\hbar$. Find the energy levels and degeneracies associated with a spin-orbit interaction term of the form

$$H_{so} = A \mathbf{L} \cdot \mathbf{S}$$

where A is a constant.

4(b) Show the stationary states for bound states in one-dimension are non-degenerate.

6

1. ***C-G coefficients required for this problem are $\langle 11; 1-1 | 11; 10 \rangle = \frac{1}{\sqrt{2}}$, $\langle 11; -11 | 11; 10 \rangle = -\frac{1}{\sqrt{2}}$, $\langle 11; 01 | 11; 11 \rangle = \frac{1}{\sqrt{2}}$, $\langle 11; 10 | 11; 11 \rangle = -\frac{1}{\sqrt{2}}$, $\langle 11; -10 | 11; 1-1 \rangle = -\frac{1}{\sqrt{2}}$ and $\langle 11; 0-1 | 11; 1-1 \rangle = \frac{1}{\sqrt{2}}$.



Indian Association for the Cultivation of Science
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End-Semester Examination

Subject: Advanced Quantum Mechanics 1

Subject Code(s): (PHS 4102)

Full marks: 50

Time allotted: 3.0 hrs

1. An electron moving freely inside a one-dimensional infinite potential box with walls at $x = 0$ and $x = a$ is initially in the ground state ($n = 1$). The right-hand wall is then instantaneously moved to $x = 4a$.
- (i) Calculate the probability of finding the electron in the ground state of the new box.
- (ii) Calculate the probability of finding the electron in the first excited state of the new box.

2 + 2

2. (a) A one-electron atom whose ground state is nondegenerate is placed in a uniform electric field in the z -direction. Obtain an approximate expression for the induced electric dipole moment of the ground state by considering the expectation value of ez with respect to the perturbed state vector computed to the first order.
- (b) The Hamiltonian $H_0 = \frac{p^2}{2m} + V(x)$ has energy eigenvalues E_n^0 . Now consider a new Hamiltonian $H = H_0 + \frac{\lambda p}{m}$. What will be the energy eigenvalues of H ?

4 + 4

3. The Hamiltonian matrix of a two-state system can be written as

$$H = \begin{pmatrix} E_1^0 & \lambda\Delta \\ \lambda\Delta & E_2^0 \end{pmatrix}$$

Clearly, the energy eigenfunctions for the unperturbed problem ($\lambda = 0$) are given by

$$\phi_1^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \phi_2^{(0)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- (a) Solve this problem exactly to find the energy eigen-functions ψ_1 and ψ_2 and the energy eigenvalues E_1 and E_2 .
- (b) Suppose the two unperturbed energies are "almost degenerate"; that is,

$$|E_1^0 - E_2^0| \ll \lambda|\Delta|.$$

Show that the exact results obtained in (a) closely resemble what you would expect by applying degenerate perturbation theory to this problem with E_1^0 set exactly equal to E_2^0 .

3 + 5

4. (a) Find an upper bound on the ground state energy of the one-dimensional infinite square well using the "triangular" trial wave-function.

$$\psi(x) = \begin{cases} Ax, & \text{if } 0 \leq x \leq \frac{a}{2} \\ A(a-x), & \text{if } \frac{a}{2} \leq x \leq a \\ 0, & \text{otherwise} \end{cases}$$

- (b) Use WKB approximation to find the allowed energies of harmonic oscillator.

4 + 3

5. (a) Directly compute the matrix elements

$$\langle j=1, m' | J^q | j=1, m \rangle$$

, Where J^q denotes the q-th spherical component of the angular momentum operator. Verify that these matrix elements satisfy Wigner-Eckart theorem, and deduce the corresponding reduced matrix elements from them.

- (b) The matrix elements of tensor operators with respect to angular momentum eigenstates satisfy

$$\langle \alpha', j' m' | T_q^{(k)} | \alpha, j m \rangle = \langle j k; m q | j k; j' m' \rangle \frac{\langle \alpha' j' || T^{(k)} || \alpha j \rangle}{\sqrt{(2j+1)}}$$

where the doublebar matrix element is independent of m, m' and q . (Wigner-Eckart theorem).

Using the theorem prove

$$\langle \alpha', j' m' | V_q | \alpha, j m \rangle = \frac{\langle \alpha', j m | \mathbf{J} \cdot \mathbf{V} | \alpha, j m \rangle}{\hbar^2 j(j+1)} \langle j m' | J_q | j m \rangle$$

where $J_{\pm 1} = \mp \frac{1}{\sqrt{2}}(J_x \pm iJ_y)$ and $J_0 = J_z$. Same goes for V_q .

4 + 4

6. Show the law of detailed balance between stimulated emission and absorption using time-dependent harmonic perturbation.

5

7. Derive the first order correction to the energy eigenvalue due to a time independent perturbing potential, from time dependent perturbation theory.

5

8. Show that, if the Hamiltonian commutes with parity, then non-degenerate energy eigenstates of the Hamiltonian are also eigenstates of parity. Further show that time reversal operator must commute with the Hamiltonian.

5



Questions for Course PHS4111

1. Write a generic function that sorts an array (Quicksort as in Computer Journal Vol 5, No 1 by C. Anthony & R. Hoare) in increasing order. Provide examples of using this function applicable to arrays of (a) integers, (b) floats or (c) strings (`std::string`). How many trials are needed for this sorting of an array of n elements (average and worst)?
2. Construct a class `ThreeVector` with cylindrical coordinates (r, ϕ, z in float): standard as well as default constructor (no input attributes); copy constructor; override methods for `=`, `==`, `+=`, `*=` (with a scalar); providing methods for dot product, cross product. Also, provide methods to return each component and a method with `ostream` (to be used to print the content of three vectors).
3. Construct a generic class of Lorentz Vector (either working in real space or momentum space) which includes the three vector class to specify the space coordinates and has a method for Lorentz transformation with an external boost factor.
4. Generate random numbers in the range between 0 and 1 which follows the Gaussian distribution, $f(x) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{x^2}{2}\right]$, with mean 0 and width 1 using the three standard methods: method of transformation, the law of large numbers, and method of rejection.
5. An experiment measures the energy of a particle where the energy is shared among 9 detector elements: $E = \sum_i C_i E_i$. The coefficients are determined by exposing the detector to particles of fixed known energy so that the measured energy matches with the incident known energy and the spread in the energy measurement from 100 such measurements has the narrowest spread (best energy resolution).
6. Illustrate the terms: Random vs Systematic uncertainties, precision vs accuracy, and determinate vs indeterminate errors. In the example of finding the mass of resonance by fitting a distribution of the number of events as a function of energy, which could be random or systematic uncertainty of the measurement and how one can estimate them?



- Lorentz Transformation

$p' (\vec{p}', \epsilon')$: Four momentum of B in rest frame of A

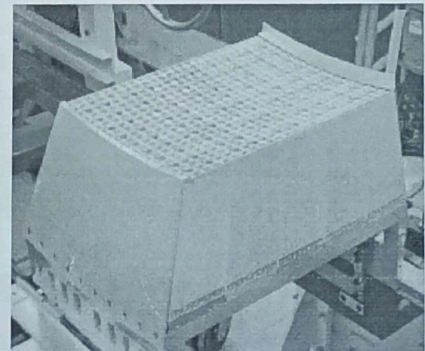
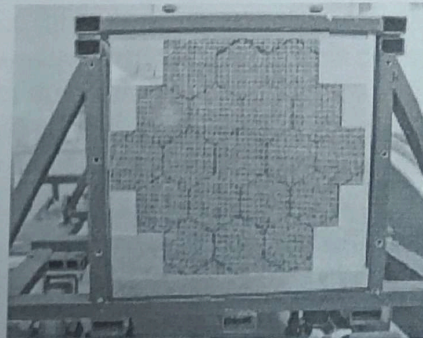
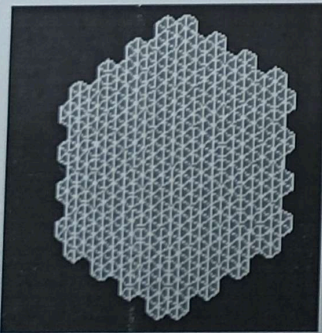
$P (\vec{P}, E)$: Four momentum of A in some frame

$p (\vec{p}, \epsilon)$: Four momentum of B in the same frame

$$\begin{aligned}\vec{p} &= \vec{p}' + \vec{\beta} \gamma \left[\frac{\gamma}{\gamma+1} \vec{\beta} \cdot \vec{p}' + \epsilon' \right] \\ \epsilon &= \gamma \left[\epsilon' + \vec{\beta} \cdot \vec{p}' \right]\end{aligned}$$

where $\vec{\beta} = \frac{\vec{p}}{E}$ and $\gamma = \frac{1}{\sqrt{1-\vec{\beta} \cdot \vec{\beta}}}$

- Calorimeter Cell: (some real-life illustrations)





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BSMS/Integrated MS-PhD Programme

End-Semester Examination-2024 (Autumn Semester)

Subject: Mathematical Methods of Physics 1
Full Marks: 50

Subject Code(s): PHS4103
Time Allotted: 3hr

The question paper has 3 pages, 7 questions; answer any five questions.

1. (a) Enunciate clearly the principle that one must use to derive Snell's law of optical refraction between two homogeneous media with refractive indices n_1 and n_2 . Show explicitly how this principle does indeed lead to Snell's law. [1+4]
(b) Consider a two dimensional flat Euclidean surface S embedded in $\mathbf{R}^4$, i.e., an Euclidean space of 4 dimensions. Is the area $A(S)$ of this surface, a vector in $\mathbf{R}^4$? Why? What about the angular momentum of a particle moving in $\mathbf{R}^4$, is this a vector in that space? Why? [3+2]
2. (a) The Indian drums known as tablas come in pairs. For right-handed tabla players, the right tabla has a different main sound than the left. Can you give a rough sketch of the frequency spectrum (sound intensity vs frequency) of the main sound of each of the two tablas, briefly explaining the difference between your sketches? [5]
(b) Can you write down the general functional structure of the radial part of the wave function of a particle in a three dimensional isotropic harmonic oscillator potential, for its n th energy level, as a function of the radial coordinate? Show how one can explicitly derive the first three terms of the polynomial part of the radial wave function, corresponding to the three lowest lying energy levels of the particle. [2+3]

3. (a) The Pauli matrices are any set of three 2×2 traceless hermitian matrices, obeying certain algebraic relations between themselves. Without using any explicit form of the matrices, state what these algebraic relations are. How does the tracelessness of the matrices emerge, from a certain property of a finite unitary rotation matrix in spin $1/2$ space ? [2+3]
- (b) For a series RLC circuit connected to an AC voltage source with frequency ω_0 , find the conditions on the circuit parameters R, L, C, ω_0 for the case that the current *leads* the source voltage, and the case that it *lags* the source voltage. Roughly sketch the corresponding phasor diagrams. [3+2]
4. (a) For exponential functions of a complex variable z , show, using the series expansion of the exponential function, that the index rule $\exp z_1 \cdot \exp z_2 = \exp(z_1 + z_2)$ holds. [5]
- (b) Derive explicitly the real and imaginary parts of the function $\sin z$ of the complex variable z , and show that it is analytic everywhere in the finite part of the complex z plane. [2+3]
5. (a) If an infinite series

$$S = a_0 + a_1(z - a) + a_2(z - a)^2 + \dots$$

converges at a point z_0 , what is the region of convergence for this series ? How would you establish your answer ? [1+4]

- (b) Consider the map $w = z^{1/4}$; is the map analytic at the origin of the w -plane ? Is w single valued for every value of z ? Why ? Sketch the precise regions in the w plane, where the special nature of this map becomes apparent. [2+3]
6. (a) Without attempting to do any explicit integration, give a brief argument as to what the value of the line integral of the quadratic function $f(z) = az^2 + bz + c$, should be around any closed curve Γ in the finite part of the complex plane ? a, b, c are complex constants. [4]
- (b) Derive explicitly the value of the line integral $\oint_{\Gamma} dz f(z)/(z - z_0)$ for the above quadratic function, if z_0 is a point in the finite part of the complex plane, within the region bounded by Γ . [6]
7. (a) How does one solve the following inhomogeneous linear second-order pde using the method of Green functions ?

$$(\nabla^2 + \lambda^2)\phi(\vec{r}) = -\rho(\vec{r}), \lambda \in \mathbf{R}$$

What additional information would you need to obtain the *most general solution* of this equation ? [1+2]

- (b) Give an explicit derivation for the Green function of this problem, justifying various choices you make to derive it. Very briefly discuss physical properties of your final answer ? [6+1]



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BSMS/Integrated MS-PhD Programme

Mid-Semester Examination-2024 (Autumn Semester)

Subject: Mathematical Methods of Physics 1
Full Marks: 25

Subject Code(s): PHS4103
Time Allotted: 2hr

Answer any five questions out of seven. Marks are within [.]

1. A particle with charge q and mass m moves in a magnetic vector potential $\vec{A}(\vec{r}, t)$ with velocity $\vec{v}$. Write down the Lagrangian for the particle, and deduce its Euler-Lagrange equation of motion explicitly using the principle of extremal action. Is this action invariant under spatial rotations for arbitrary $\vec{A}(\vec{r}, t)$? Why? [3+2]
2. Consider three arbitrary linearly independent, not necessarily orthogonal, vectors in $\mathbf{R}_3$, $\vec{A}, \vec{B}, \vec{C}$. Construct an orthonormal basis in $\mathbf{R}_3$ in terms of $\vec{A}, \vec{B}, \vec{C}$. Expand $\vec{A}, \vec{B}, \vec{C}$ in the constructed orthonormal basis. [3+2]
3. Expand the function $f(x) = x$ for $0 < x < l/2$, $f(x) = 1 - x$ for $l/2 < x < l$ in a Fourier series with the first three non-trivial Fourier coefficients shown. [5]
4. In a spherical polar coordinate basis, the angular momentum operators of quantum mechanics have the differential operator representations $\hat{L}_z = -i\hbar\partial_\phi$, $\hat{L}_\pm = -i\hbar e^{\pm i\phi}[\pm i\partial_\theta - \cot\theta\partial_\phi]$. Using properties of these operators, deduce the angular part of any p -wave function of a hydrogen atom. Give a rough sketch of the shape of this angular wave function, clearly showing the 'poles'. What multipole moment in electrostatics would this correspond to? [3+1+1]
5. Consider a rank 3 tensor in $\mathbf{R}_3$. Choose appropriate vectors in $\mathbf{R}_3$ to construct a scalar in $\mathbf{R}_3$, with this rank 3 tensor. Express the scalar you have constructed in terms of the Cartesian components of the rank 3 tensor as well as those of the vectors chosen by you. [1+4]

6. Obtain the Green function $\tilde{G}_\lambda(\vec{k})$, i.e., the Fourier transform of the Green function $G_\lambda(\vec{r} - \vec{r}')$, appropriate to the linear second order pde $(\nabla^2 - \lambda^2)\phi(\vec{r}) = \rho(\vec{r})$ for $\lambda \in \mathbf{R}$. Perform the angular part of the inverse Fourier integral, leaving the answer for $G_\lambda(\vec{r} - \vec{r}')$ in terms of the one dimensional radial integral over $|\vec{k}|$. What do you expect $\lim_{\lambda \rightarrow 0} G_\lambda(\vec{r} - \vec{r}')$ to be? From this information, can you guess the full form of $G_\lambda(\vec{r} - \vec{r}')$ upto overall multiplicative constants which you do not have to determine? [2+1+1+1]
7. Suppose we come to know that a three dimensional wave must decay and vanish at spatial infinity $|\vec{r}| \rightarrow \infty$, because there is no source at spatial infinity to supply energy to the wave. Thus, the function $f(\vec{r}, t) = f_0 \exp i(\omega t - \vec{k} \cdot \vec{r})$ for constant amplitude f_0 , which represents a plane wave solution, cannot be physical. How should f_0 change with $|\vec{r}|$, such that the wave decays to zero at spatial infinity? Can you show explicitly in this case that the wave function satisfies the three dimensional wave equation? [2+3]



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Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course
Mid-Semester Examination-Autumn 2024

Subject: Classical mechanic

Subject Code(s): PHS 4101/AIS 4110

Full Marks: 25

Time Allotted: 2 h

Answer Question No. 1 (Q 1) and Any *One* from the Rest

Q 1. (a) (i) State and (ii) prove the fundamental lemma of the calculus of variation (2 + 3)

(b) Deduce the Euler-Lagrange equation for a mechanical system with N generalized coordinates. (5)

(c) Let's consider a vertical X - Y plane uniform gravitational acceleration (g) acting along the negative Y -direction. Let there be a fixed, frictionless Archimedean spiral wire on the vertical X - Y plane, given by the equation:

$$y = \pm x \tan \sqrt{x^2 + y^2}$$

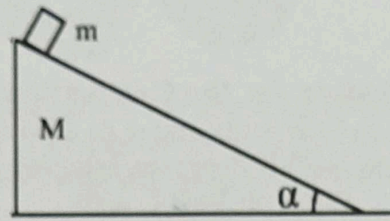
The wire is passing through a bead of mass m which slides frictionlessly through the wire.

- (i) Assign generalized coordinate(s) for the bead, clearly mentioning its/their range(s).
(ii) Deduce the Euler-Lagrange equation(s) of motion for the bead.

[Hint: The equation of an Archimedean spiral is simpler in plane-polar coordinates] (1+4)

Q 2. (a) A straight, thick, rigid rod passes through the centers of two cubes. For each cube, it passes perpendicularly through a pair of opposite faces. The cubes and the rod are allowed to rotate with respect to each other, but the cubes are not allowed to slide along the rod. Explain the number of degrees of freedom the system will have. (3)

(b) Deduce Euler-Lagrange equation for a system with N degrees of freedom under K holonomic constraints using the method of Lagrange's multiplier. You can use results from your answer to 1(b). (3)



(c) Consider a small block of mass m is sliding on the inclined face of a wedge of mass M (as shown in the Fig. above). The angle of inclination is α . There is no friction between the small block and the surface of the wedge, but there is friction between the ground and the wedge of mass M . Find the reaction force on the block of mass m , due to the constraint on it to stay on the surface of the wedge in the limit $M \rightarrow \infty$ using the method of Lagrange's multiplier. Gravity works vertically downward.

(4)

Q 3. (a) Define holonomic and semi-holonomic constraints.

(1+1)

(b) Show that a total time derivative of a function of generalized coordinates and time is added to a Lagrangian, then it is still a valid Lagrangian.

(2)

(c) Show that a Lagrangian is a scalar under transformation in the configuration space.

(3)

(d) Suppose a Lagrangian L for a system of particles is explicitly time-independent. Then prove that the following quantity is conserved.

$$L - \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i}$$

Here,

$q_i \rightarrow$ generalized coordinates,

$\dot{q}_i \rightarrow$ generalized velocities.

(3)



Indian Association for the Cultivation of Science

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Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's Program/PhD Course

End-Semester Examination-Autumn 2024

Subject: Classical Mechanics
Full Marks: 50

Subject Code(s): PHS 4101/AIS 4110
Time Allotted: 3 h

Answer Q1 and any two from the rest

Q1. (a) Deduce Hamilton's canonical equation of motion for a system with degrees of freedom interacting via a position-dependent potential. (5)

(b) Deduce Euler-Lagrange equation of motion for N interacting degrees of freedom under k Holonomic constraints. (5)

(c) (i) Define canonical transformation.

(ii) Show that point transformations (transformations that involve only generalized coordinates and time and not any other variable) are always canonical. (2+3)

(d) (i) Explain whether Newton's laws of motion hold for the motion of two interacting charged particle moving in each other's field (Hint: It is enough to construct a simple situation where at least one of the laws is violated).

(ii) It is believed that action of all fundamental forces can be described using variational principle. Why then one cannot describe viscous force within Lagrangian formalism? (4+1)

Q2. (a) The Fermat principle states that light always propagates along a path that takes the minimum amount of time. Consider an optical medium with an index of refraction given by

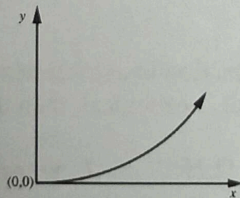
$$n(x; y) = n_0(1 + k y), \quad (k \text{ is a constant}).$$

Recall that the speed of light in a medium with index n is given by $v = c/n$.

Find **(i)** The differential Equation for the path.

(ii) Solve it to find the function that describes the path of light in this medium.

(iii) Determine a specific equation for the path of a laser beam that initially starts at the origin propagating in the x direction, as shown in the figure.



(3+3+2)

- (b) (i) Deduce the equation of motion, and
(ii) Find the general solution the equation of motion for a system with Lagrangian

$$\begin{aligned} L &= T - V, \text{ where} \\ T &= \frac{1}{2}m\dot{q}^2 + e^qt^2\dot{q} \\ V &= \frac{1}{2}kq^2 - 2e^qt \end{aligned} \quad (3+2)$$

(c) A loop made of a circular frictionless wire of Mass M is mounted vertically on a frictionless bearing wheel such that the wheel can rotate without friction about its vertical diameter. The loop passes through a bid of mass m that can slide along the loop without friction. The wheel is given an initial velocity and it starts rotating, while the bid starts to slide down the loop. Assuming there is no air resistance and gravity acts vertically downward ($-z$ direction), explain if there will be any change in the z -component of the angular momentum of the wire loop with time? (2)

3(a) Determine whether the following transformation ($q, p \leftrightarrow Q, P$) is canonical: $P = p \cos(q)$, $Q = p \sin(q)$ Assuming (q, p) are canonical. (4)

(b) Suppose $H(\mathbf{q}, \mathbf{p})$ is a Hamiltonian that satisfies canonical equations of motion with respect to $\mathbf{q}$ and $\mathbf{p}$ ($2N$ phase-space coordinates $q_1, q_2, q_N \dots p_1, p_2, p_N$). Determine a condition for a transformation ($\mathbf{q}, \mathbf{p} \leftrightarrow \mathbf{Q}, \mathbf{P}$) to be canonical starting from the fundamental definition of a canonical transformation. (5)

(c) Consider a simple Harmonic oscillator Hamiltonian

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2q^2$$

Generate the Canonical transformation ($q, p \leftrightarrow Q, P$) using the generator

$$F_1(q, Q) = \frac{1}{2}m\omega q^2 \cot Q$$

Identify the symmetry of the system in the transformed space. (4+2)

4(a) Deduce the equation of motion of an observable of a system executing Hamiltonian dynamics, using Poisson's bracket formulation. (4)

(b) Show that if J is the Jacobian of a transformation in a $2N$ dimensional phase space from one set of phase space coordinates to another, and the former set is canonical, then the transformation will be canonical if

$$\begin{aligned} \Omega J \Omega^{-1} &= J, \text{ where} \\ \Omega &= \begin{bmatrix} 0_{N \times N} & \mathcal{I}_{N \times N} \\ -\mathcal{I}_{N \times N} & 0_{N \times N} \end{bmatrix}. \end{aligned}$$

Here Ω is a $2N \times 2N$ Symplectic matrix as shown above. (3)

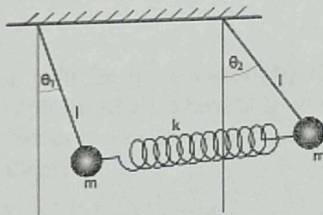
(c) If $A(\mathbf{q}, \mathbf{p})$ and $B(\mathbf{q}, \mathbf{p})$ are two dynamical quantities with no explicit time dependence, and are conserved under the dynamics of the canonical phase space coordinates $\mathbf{q}$ and $\mathbf{p}$, then show $\{A, B\}$ is also conserved. (2)

(d) Show that if fundamental Poisson brackets remain invariant under a transformation, then all Poisson Bracket remain invariant under that transformation. (3+3)

(5)(a) For a system of particles attached to each other with massless springs in equilibrium, show that in the absence of any dissipation, the frequencies of small oscillation of the normal modes about the equilibrium are always positive. Start your deduction from the Lagrangian. [Hint: You can use Cartesian coordinates if you want] (5)

(b) Show that the above system can be mapped to a set of independent simple harmonic oscillators. (4)

(c) Two identical simple pendulums, each made of a rigid rod of length ℓ and a bob of mass m , are hanging from the same horizontal bar and oscillating along a vertical plane. The two bobs are coupled with a spring with spring constant k . The spring remains horizontal in its natural configuration (no stretching or compression) when the two pendulums are at their equilibrium positions (see the Fig below). Find the Lagrange's equation of motion and normal frequencies of the system.



(3+3)

(6) (a) State and prove Noether's theorem establishing a relation between continuous symmetries of a system and relevant conservation laws (it will be sufficient to prove the version which covers spatial symmetries). (2+3)

(b) A particle of mass m is in a potential $U(\rho, \phi, z)$ such that

$$U(\rho, \phi, z) = U(\rho, a\phi + z),$$

where a is a constant, and (ρ, ϕ, z) are the standard cylindrical polar coordinates of the particle. Identify the continuous symmetry and find the associated Noether's charge (conserved quantity). (2+3)

(c) Show that the equations of motion for a circular wheel rolling on a horizontal 2-dimensional plane without slipping are non-integrable. (5)



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Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course
Mid-Semester Examination-Autumn 2024

Subject: Electronics (Theory) *Subject Code(s): PHS 4104/AIS 4113/PHD117*
Full Marks: 25 *Time Allotted: 2 h*

Semiconductor Device Physics (13 Marks)

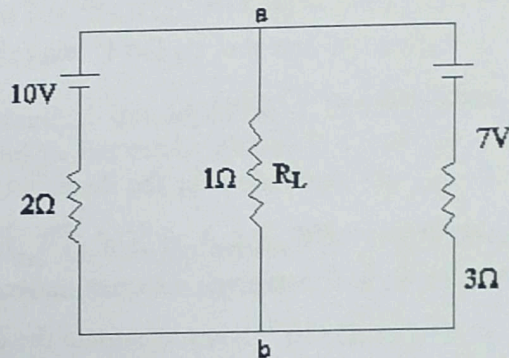
1. The resistivity of a semiconductor decreases from $1 \Omega\cdot\text{cm}$ at 300 K to $0.1 \Omega\cdot\text{cm}$ at 350 K. Assuming that the resistivity $\rho(T)$ is inversely proportional to the carrier concentration and that $n_i(T) \propto T^{3/2} \exp(-E_g/2k_B T)$, determine the bandgap E_g of the semiconductor. (3 marks)
2. Derive the expression for the depletion width (W) of a PN junction. Calculate the depletion width for a silicon PN junction at 300 K: Donor concentration, $N_D = 10^{16} \text{ cm}^{-3}$, Acceptor concentration, $N_A = 10^{15} \text{ cm}^{-3}$, Assume $V_{bi} = 0.7 \text{ V}$ and $\epsilon_s = 11.7 \times 8.85 \times 10^{-14} \text{ F/cm}$, Compute the depletion width W for the given parameters. (4 marks)
3. Explain how the Fermi level shifts with respect to the intrinsic level as a function of the donor concentration (N_D) and temperature (T). Derive the expression for the Fermi level E_F in terms of the intrinsic Fermi level E_i , the donor concentration, and the intrinsic carrier concentration n_i . Calculate the shift in the Fermi level for an n-type silicon semiconductor at 300 K. Given: Donor concentration, $N_D = 10^{16} \text{ cm}^{-3}$, Intrinsic carrier concentration for silicon at 300 K, $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, Boltzmann constant, $k_B = 8.617 \times 10^{-5} \text{ eV/K}$, Calculate the shift in the Fermi level E_F with respect to the intrinsic level E_i . (3 marks)
4. Explain what a degenerate semiconductor is, and how does it differ from a non-degenerate semiconductor? Draw the Energy Band Diagram for a Degenerate Semiconductor: Illustrate the energy band diagram of a degenerate semiconductor, including, the density of states (DOS) for a 3D material, the Fermi function, Carrier

concentration distribution with energy. Label the conduction band, valence band, Fermi level, and the associated energy states. (3 marks)

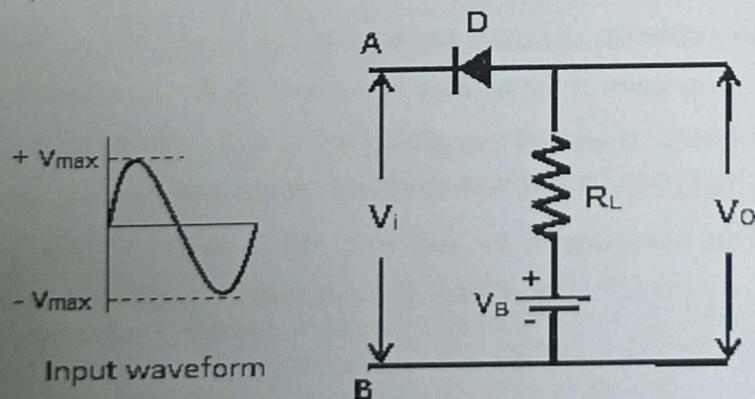
Circuit theory (12 Marks)

5. A transmission line has the following per unit length parameters: $L=0.2\mu\text{H/m}$, $C=300\text{pF/m}$, $R=5\Omega/\text{m}$ and $G=0.01\text{S/m}$. Calculate the propagation constant and the characteristic impedance (in rectangular form) of this line at 500MHz . (3 marks)
6. Seawater at a frequency $f = 9 \times 10^2 \text{ Hz}$ has permittivity $\epsilon = 80\epsilon_0$ and resistivity $\rho = 0.25 \Omega\text{m}$. Imagine a parallel plate capacitor is immersed in seawater and is driven by an alternating voltage source $V(t) = V_0 \sin(2\pi ft)$. Then, the conduction current density becomes 10^x times the displacement current density after time $t=1/800 \text{ s}$. Find the value of x . $[(1/4\pi\epsilon_0=9 \times 10^9 \text{ Nm}^2/\text{C}^2)]$ (3 marks)

7. a) Find the individual battery currents.
b) The current through the load.
c) The voltage across the load.
[b is grounded]
(3 marks)



8. Draw the output waveform (in the same scale as input) considering the diode as ideal. (3 marks)





Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course
End-Semester Examination-Autumn 2024

Subject: Electronics
Full Marks: 50

Subject Code(s): PHS 4104/ AIS 4113/PHD 117
Time Allotted: 3 h

Semiconductor Device Physics

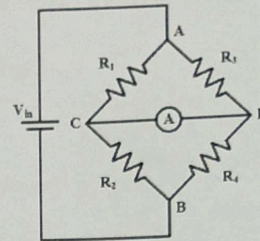
1. What is compensation doping in semiconductors? A silicon sample is doped with donor atoms at a concentration of $N_D = 6 \times 10^{16} \text{ cm}^{-3}$ and acceptor atoms at $N_A = 3 \times 10^{16} \text{ cm}^{-3}$, discuss whether the semiconductor behaves like an n-type or p-type material under these conditions. **(3 marks)**
2. Derive the expression for the contact potential developed in a p-n junction at equilibrium. Explain how this potential prevents further carrier motion and stabilizes the junction. **(3 marks)**
3. Using the Shockley diode equation, derive the relationship between the forward bias current and the applied voltage in a p-n junction. Include the significance of the ideality factor. **(3 marks)**
4. A Schottky diode with a metal-semiconductor junction (n-Si and Au) has a barrier potential of 0.83 eV. Draw the energy band diagram of the metal-semiconductor junction in equilibrium (Work function of Au: $\Phi_{Au} = 5.1 \text{ eV}$, electron affinity of Si: $\chi_{Si} = 4.05 \text{ eV}$, and energy gap of Si: $E_g = 1.1 \text{ eV}$). If the reverse bias voltage is 50 V, calculate the width of the depletion region. Assume the doping concentration of the semiconductor to be $N_d = 10^{15} \text{ cm}^{-3}$. **(4 marks)**
5. Discuss the working principle of a tunnel diode. Explain how negative differential resistance arises in its I-V characteristics. Include a labeled energy band diagram. **(4 marks)**
6. What is the open-circuit voltage (V_{oc}) in a solar cell? Explain its significance in determining solar cell performance. Derive an expression for (V_{oc}) in terms of the photon flux (ϕ_{ph}) and reverse saturation current (I_s). Include the assumptions made in your derivation and discuss the physical meaning of the result. **(4 marks)**
7. Explain the working principle of a double heterostructure (DH) LED in detail, emphasizing how the double heterostructure improves carrier confinement, photon confinement, and overall light emission efficiency compared to a standard p-n junction LED. Use appropriate diagrams to illustrate the following: **(4 marks)**

- (i) Draw the energy band diagram for a DH LED with:
- $E_g = 2.0 \text{ eV}$ (n^+ -doped AlGaAs cladding layer)
 - $E_g = 1.4 \text{ eV}$ (p-GaAs active layer)
 - $E_g = 2.0 \text{ eV}$ (p-AlGaAs cladding layer)
- (ii) Show the movement of electrons and holes, the recombination process in the active layer, and the photon emission mechanism.

Electronic circuits

8. In a typical strain-gauge-based Wheatstone bridge:

(a) The strain gauge replaces one resistor, say R_1 , and its resistance changes by ΔR due to strain. The other resistors (R_2, R_3, R_4) are equal to R_0 (the initial resistance of the strain gauge). Consider $\Delta R \ll R_0$. Prove that the output voltage V_{out} (voltage between C and D in the figure) of the above Wheatstone bridge would be close to $(4R_0/\Delta R)$ times the input voltage (V_{in}).



(b) A strain gauge with a gauge factor $GF=2.1$ and an initial resistance $R_0=120 \Omega$ is connected in a Wheatstone bridge circuit. The bridge is powered by a DC voltage source of $V_{in}=10 \text{ V}$. The strain gauge is mounted on a steel beam with a Young's modulus $E=200 \text{ GPa}$ and a cross-sectional area $A=25 \text{ mm}^2$. A tensile force $F=10,000 \text{ N}$ is applied to the beam.

1. Calculate the strain in the beam.
2. Determine the change in resistance ΔR of the strain gauge.
3. Calculate the output voltage V_{out} of the Wheatstone bridge when the strain gauge is part of the circuit. Assume the other three resistors in the bridge are 120Ω .

(10 marks)

9. Solve the following problem using the concept of Laplace Transform:

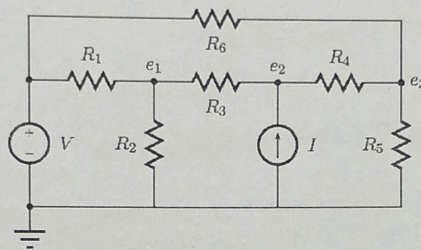
A series RLC circuit consists of: A resistor $R = 10 \Omega$, an inductor $L = 1 \text{ H}$, a capacitor $C = 0.01 \text{ F}$, connected in series with a voltage source $V(t)$, where: $V(t) = 10 \sin(2t) \text{ V}$ for $t \geq 0$.

1. Write the differential equation governing the circuit.
2. Transform the equation into the Laplace domain.
3. Solve for the current $I(s)$ in the Laplace domain.
4. Comment on the different terms of time-domain current $i(t)$ guessing the partial fraction decomposition of $I(s)$ and utilizing Laplace transform of $\sin(t)$, $\cos(t)$, $\exp(at)\cos(bt)$, $\exp(at)\sin(bt)$. No need to do the exact calculation.

(10 marks)

$$I = \int_0^t e^{-st} \sin 2t \, dt$$

10. Using the node method, develop a set of simultaneous equations for the network shown below that can be used to solve for the three unknown node voltages in the network. Express these equations in the matrix form $G [e] = S$, where G ($G = R^{-1}$) is a matrix of conductance terms and S is a 3×1 vector of terms involving the sources. "e" denotes the node voltage. You can verify the result using the KCL/KVL. You need not solve the set of equations for the node voltages. (5 marks)





Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)

Quiz

Subject: Advanced Quantum Mechanics 1

Subject Code(s): (PHS 4102)

Full marks: 15

Time allotted: 1.5 hrs

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1. A particle of mass m is constrained to move in a circle of radius a . Calculate the energy eigen values and eigen functions of the particle.
 2. Obtain the unitary operator corresponding to a finite translation and show that the conservation of the total momentum of an isolated system results from the invariance of its Hamiltonian under translations.
 3. Obtain the selection rules (i) $m_1 + m_2 = m$ and (ii) $|j_1 - j_2| \leq j \leq j_1 + j_2$ for the Clebsch-Gordan (CG) coefficients $\langle j_1 j_2; m_1, m_2 | j m \rangle$

5+5+5